

DHRUV KUMAR

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RESEARCH INTERESTS	Large Language Models, AI Agents, Human-Computer Interaction, Computing Education	
CURRENT AFFILIATION	BITS Pilani , Pilani, India <i>Assistant Professor</i>	May 2024 - Present
EDUCATION	University of Minnesota , Twin Cities, United States <i>Ph.D. in Computer Science</i> [CGPA: 4.0 / 4.0]	Sep 2017 - Mar 2022
	Birla Institute of Technology and Science (BITS), Pilani, India <i>B.E. (Hons.) Computer Science</i> [CGPA: 9.92 / 10.0] • Rank 1 in the class of 2014 of Computer Science, comprising 120 students. • Rank 4 in the entire class of 2014, comprising 800 students.	Aug 2010 - May 2014
PREVIOUS WORK EXPERIENCE	IIIT Delhi , New Delhi, India <i>Assistant Professor</i>	Aug 2022 - May 2025
	Microsoft Research , Bengaluru, India <i>Post-Doctoral Researcher</i>	Mar 2022 - Jul 2022
	Google Cloud , Sunnyvale, USA <i>Summer Ph.D. Intern</i> <i>Technology and Strategy</i>	May 2019 - Aug 2019
	Goldman Sachs , Bengaluru, India <i>Software Developer, Investment Management Division</i>	Nov 2014 - Apr 2016
HONORS AND AWARDS	Teaching Excellence Award for AY 2023-24, IIIT Delhi Teaching Excellence Award for AY 2022-23, IIIT Delhi 3M Science and Technology Fellowship, UMN Graduate School Student Travel Grant, ATC 2019 Student Travel Grant, SIGMETRICS 2019 Student Travel Grant, OSDI 2018 Merit Scholarship (Top 10 among 800 students), BITS Pilani Research Fellowship (Outstanding undergraduate thesis), BITS Pilani	Sep 2024 Sep 2023 Aug 2017 - Jun 2021 Jul 2019 Jun 2019 Oct 2018 Aug 2010 - May 2014 Jan 2014 - May 2014
SPONSORED PROJECTS	AI for River-Climate Modeling [Amount: INR 25,00,000] Funding Agency: The Nature Conservancy	Sep 2024 - Aug 2025
	New Faculty Seed Grant [Amount: INR 20,00,000] Funding Agency: BITS Pilani	Nov 2024 - Oct 2026
	Generative AI in Education Grant [Amount: USD 30,000] Funding Agency: WISE-IIE Higher Education Consortium	Jul 2024 - May 2025

Translational Research Grant [Amount: INR 99,40,000] Funding Agency: iHub Anubhuti-IIITD Foundation	Mar 2024
Google Cloud Education Grant [Amount: INR 12,50,000] Funding Agency: Google Cloud	Jan 2024 - Dec 2024
Startup Research Grant [Amount: INR 22,35,000] Funding Agency: DST-SERB	Oct 2023
Initiation Research Grant [Amount: INR 10,00,000] Funding Agency: IIT Delhi	Aug 2022

TEACHING
EXPERIENCE

Programming Languages (M24), Object-Oriented Programming (M24), Compiler Construction (W25), Large Language Models (M24, W25) at BITS Pilani

Operating Systems (M23), Cloud Computing (M22), Distributed Systems (W23, W24) at IIT Delhi

Graduate Teaching Assistant for Operating Systems (Fall 2020) at UMN-TC, USA

SELECTED
PUBLICATIONS

Vikas Hassija, Balamurugan Palanisamy, Arpita Chatterjee, Arpita Mandal, Debanshi Chakraborty, Amit Pandey, GSS Chalapathi, and **Dhruv Kumar**. *Transformers for Vision: A Survey on Innovative Methods for Computer Vision*, in **IEEE Access**, 2025. [\[Link\]](#)

Harshita Goyal, Garima Garg, Prisha Mordia, Veena Ramachandran, **Dhruv Kumar**, and Jagat Sesh Challa. *The Impact of Large Language Models on K-12 Education in Rural India: A Thematic Analysis of Student Volunteer's Perspectives*, in **arXiv preprint arXiv:2505.03163**, 2025. [\[Link\]](#)

Aditya Pathak, Rachit Gandhi, Vaibhav Uttam, Devansh, Yashwanth Nakka, Aaryan Raj Jindal, Pratyush Ghosh, Arnav Ramamoorthy, Shreyash Verma, Aditya Mittal, Aashna Ased, Chirag Khatri, Jagat Sesh Challa, and **Dhruv Kumar**. *Rubric Is All You Need: Enhancing LLM-based Code Evaluation With Question-Specific Rubrics*, in **ICER 2025**. [\[Link\]](#)

Harshal Akolekar, Piyush Jhamnani, Vikash Kumar, Vinay Tailor, Aditya Pote, Ankit Meena, Kamal Kumar, Jagat Sesh Challa, and **Dhruv Kumar**. *The role of generative AI tools in shaping mechanical engineering education from an undergraduate perspective*, in **Nature Scientific Reports**, 2025. [\[Link\]](#)

Raj Gupta, Harshita Goyal, Apurv Mehra, Sanchit Sharma, Kashish Mittal, Jagat Sesh Challa, and **Dhruv Kumar**. *Saksham AI: Advancing AI-Assisted Coding Education for Engineering Students in India Through Socratic Tutoring and Comprehensive Feedback*, in **arXiv preprint arXiv:2503.12479**, 2025. [\[Link\]](#)

Chavvi Kirtani, Madhav Krishan Garg, Tejash Prasad, Tanmay Singhal, Murari Mandal, and **Dhruv Kumar**. *ReviewEval: An Evaluation Framework for AI-Generated Reviews*, in **arXiv preprint arXiv:2502.11736**, 2025. [\[Link\]](#)

Utkarsh Arora, Anupam Garg, Aryan Gupta, Samyak Jain, Ronit Mehta, Rupin Oberoi, Prachi, Aryaman Raina, Manav Saini, Sachin Sharma, Jaskaran Singh, Sarthak Tyagi, and **Dhruv Kumar**. *Analyzing LLM Usage in an Advanced Computing Class in India*, in **Proceedings of the 27th Australasian Computing Education Conference**, 2025. [\[Link\]](#)

Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, and **Dhruv Kumar**. *DebateBench: A Challenging Long Context Reasoning Benchmark for Large Language Models*, in **NAACL SRW**

2025. [\[Link\]](#)

Shreya Bhatia, Tarushi Gandhi, and **Dhruv Kumar**, Pankaj Jalote. *System Test Case Design from Requirements Specifications: Insights and Challenges of Using ChatGPT*, in **arXiv preprint arXiv:2412.03693**, 2024. [\[Link\]](#)

Anishka IIITD, Diksha Sethi, Nipun Gupta, Shikhar Sharma, Srishti Jain, Ujjwal Singhal, and **Dhruv Kumar**. *TAMIGO: Empowering Teaching Assistants Using LLM-Assisted Viva and Code Assessment in an Advanced Computing Class*, in **arXiv preprint arXiv:2407.16805**, 2024. [\[Link\]](#)

Shubhang Shekhar Dvivedi, Vyshnav Vijay, Sai Leela Rahul Pujari, Shoumik Lodh, and **Dhruv Kumar**. *A Comparative Analysis of Large Language Models for Code Documentation Generation*, in **Proceedings of the 1st ACM International Conference on AI-Powered Software**, 2024. [\[Link\]](#)

Shikhar Sharma, Manas Mhasakar, Apurv Mehra, Utkarsh Venaik, Ujjwal Singhal, **Dhruv Kumar**, and Kashish Mittal. *Comuniqua: Exploring Large Language Models for Improving English Speaking Skills*, in **Proceedings of the 7th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies**, 2024. [\[Link\]](#)

Harshal Dev, Raj Gupta, and **Dhruv Kumar**. *From Cash to Cashless: UPI's Impact on Spending Behavior among Indian Users*, in **Extended Abstracts of the CHI Conference on Human Factors in Computing Systems**, 2024. [\[Link\]](#)

Shreya Bhatia, Tarushi Gandhi, and **Dhruv Kumar**, Pankaj Jalote. *Unit Test Generation Using Generative AI: A Comparative Performance Analysis of Autogeneration Tools*, in **Proceedings of the 1st International Workshop on Large Language Models for Code**, 2024. [\[Link\]](#)

I Joshi, R Budhiraja, HD Akolekar, JS Challa, and **Dhruv Kumar**. *"It's not like Jarvis, but it's pretty close!" – Examining ChatGPT's Usage among Undergraduate Students in Computer Science*, in **ACE 2024**. [\[Link\]](#)

I Joshi, R Budhiraja, PD Tanna, L Jain, M Deshpande, A Srivastava, S Rallapalli, HD Akolekar, JS Challa, **Dhruv Kumar**. *"With Great Power Comes Great Responsibility!": Student and Instructor Perspectives on the influence of LLMs on Undergraduate Engineering Education*, in **arXiv 2023**. [\[Link\]](#)

H Dev, J Kadia, MO Ataullah, S Mitra, **Dhruv Kumar**, HD Akolekar. *ChatGPT in the Classroom: An Analysis of Its Strengths and Weaknesses for Solving Undergraduate Computer Science Questions*, in **SIGCSE TS 2024**. [\[Link\]](#)

Dhruv Kumar, S Ahmad, A Chandra, R Sitaraman. *AggFirstJoin: Optimizing Geo-Distributed Joins using Aggregation-Based Transformations*, accepted in **IEEE/ACM CCGrid 2023**. [\[Link\]](#) [\[Best Paper Award\]](#)

J Wolfrath, N Sreekumar, **Dhruv Kumar**, Y Wang, A Chandra. *HACCS: Heterogeneity-Aware Clustered Client Selection for Accelerated Federated Learning*, in **IEEE IPDPS 2022**. [\[Link\]](#)

Dhruv Kumar, S Ahmad, A Chandra, R Sitaraman. *AggNet: Cost-Aware Aggregation Networks for Geo-distributed Streaming Analytics*, in **ACM/IEEE SEC 2021**. [\[Link\]](#)

Dhruv Kumar, J Li, A Chandra, R Sitaraman. *A TTL-based Approach for Data Aggregation in Geo-Distributed Streaming Analytics*, in **ACM SIGMETRICS 2019**. [\[Link\]](#)

OTHER
PUBLICATIONS

S Rallapalli, A Jain, **Dhruv Kumar**. *Cloud-based neuro-fuzzy hydro-climatic model for water quality assessment under uncertainty and sensitivity*, in **Environmental Science and Pollution Research**, Springer, 2022. [\[Link\]](#)

R Srinivas, M Drewitz, J Magner, A P Singh, **Dhruv Kumar**, Y B Katpatal. *Simulating Landscape Hydrologic Connectivity in a Precise Manner Using Hydro-Conditioning*, in **Advances in Computational Modeling and Simulation**, Springer, 2022. [\[Link\]](#)

Dhruv Kumar, J Wolfrath, A Chandra, R Sitaraman. *Towards WAN-Aware Join Sampling over Geo-Distributed Data*, in **ACM EdgeSys 2022**. [\[Link\]](#)

Y Wang, J Wolfrath, N Sreekumar, **Dhruv Kumar**, A Chandra. *Accelerated Training via Device Similarity in Federated Learning*, in **ACM EdgeSys 2021**. [\[Link\]](#)

Y Wang, **Dhruv Kumar**, A Chandra. *Exploiting Data Heterogeneity for Performance and Reliability in Federated Learning*, Poster in **ACM/IEEE SEC 2020**. [\[Link\]](#)

Dhruv Kumar, A A Ramkumar, R Sindhu, A Chandra. *DeCaf: Iterative Collaborative Processing over the Edge*, in **USENIX HotEdge 2019**. [\[Link\]](#)

P Goyal, JS Challa, **D Kumar**, A Bhat, N Goyal, Sundar B. *Grid-R-tree: A data structure for efficient neighborhood and nearest neighbor queries in data mining*, in **JDSA Springer 2020**. [\[Link\]](#)

Dhruv Kumar, J Li, A Chandra, R Sitaraman. *A TTL-based Approach for Data Aggregation in Geo-Distributed Streaming Analytics*, in **ACM SIGMETRICS 2019**. [\[Link\]](#)

Dhruv Kumar, J Li, A Chandra, R Sitaraman. *TTL-based Approach for Data Aggregation in Geo-Distributed Streaming Analytics*, Poster in **OSDI 2018**. [\[Link\]](#)

Dhruv Kumar, P Goyal, N Goyal. *An Efficient method for Batch Updates in OPTICS Cluster Ordering*, in **IJDATS 2018**. [\[Link\]](#)

P Goyal, S Kumari, A Sood, **Dhruv Kumar**, Sundar B, and N Goyal. *Exact, Fast and Scalable Parallel DBSCAN for Commodity Platforms*, in **ICDCN 2017**. [\[Link\]](#)

P Goyal, S Kumari, S Sharma, **Dhruv Kumar**, V Kishore, Sundar B, and N Goyal. *A fast, Scalable SLINK Algorithm for Commodity Cluster Computing Exploiting Spatial Locality*, in **HPCC 2016**. [\[Link\]](#)

P Goyal, S Kumari, **Dhruv Kumar**, Sundar B, N Goyal, S Islam, and JS Challa. *Parallelizing OPTICS for Commodity Clusters* in **ICDCN 2015**. [\[Link\]](#)

P Goyal, S Kumari, **Dhruv Kumar**, Sundar B, and N Goyal. *Parallelizing OPTICS for multicore systems* in **ACM COMPUTE 2014**. [\[Link\]](#)

SERVICE

Reviewed Papers for Elsevier JPDC, IPDPS (2020, 2019), Big Data (2020, 2019), IISWC 2020, ICDCS 2020